

```

45.     # Extract train and test data
46.     train_data = full_data[(full_data.index >= current_start) & (full_data.index < train_end)]
47.     test_data = full_data[(full_data.index >= train_end) & (full_data.index < test_end)]
48.
49.     # Ensure we have enough data
50.     if len(train_data) < 50 or len(test_data) < 20:
51.         current_start = train_end
52.         continue
53.
54.     # Optimize parameters on training data
55.     best_params = None
56.     best_sharpe = -float('inf')
57.
58.     for short_ma in param_grid['short_ma']:
59.         for long_ma in param_grid['long_ma']:
60.             # Skip invalid combinations
61.             if short_ma >= long_ma:
62.                 continue
63.
64.             # Calculate indicators
65.             train_data['SMA_Short'] = train_data['Adj Close'].rolling(window=short_ma).mean()
66.             train_data['SMA_Long'] = train_data['Adj Close'].rolling(window=long_ma).mean()
67.
68.             # Generate signals
69.             train_data['Signal'] = 0
70.             train_data.loc[train_data['SMA_Short'] > train_data['SMA_Long'], 'Signal'] = 1
71.
72.             # Run backtest
73.             backtest_result = backtest_strategy(train_data)
74.
75.             # Evaluate results
76.             metrics = evaluate_backtest(backtest_result)
77.
78.             # Check if this is the best parameter set so far
79.             if metrics['Sharpe Ratio'] > best_sharpe:
80.                 best_sharpe = metrics['Sharpe Ratio']
81.                 best_params = (short_ma, long_ma)
82.

```